

Invasion of the Home Humanoid Robots [C1]

Un robot maggiordomo che fa le pulizie e piega il bucato: non è un film di fantascienza, è la realtà. Conosciamo Neo, l'umanoide con accento scandinavo che ci aiuterà con le faccende domestiche.

On a recent morning, I knocked on the front door of a handsome two-story home in Redwood City, California. Within seconds, the door was opened by a faceless robot dressed in a beige bodysuit that [clung tight](#) to its [trim waist](#) and long legs. This svelte humanoid greeted me with what seemed to be a Scandinavian accent, and I offered to shake hands. As our palms met, it said: "I have a [firm grip](#)." When the home's owner, a Norwegian engineer named Bernt Børnich, asked for some bottled water, the robot turned, walked into the kitchen and opened the refrigerator with one hand. Artificial intelligence is already driving cars, writing essays and writing computer code. Now humanoids, machines built to look like humans and powered by AI, are [poised](#) to move into our homes so they can help with the [daily chores](#). Børnich is CEO and founder of a startup called 1X. Before the end of the year, his company hopes to put his robot, Neo, into more than one hundred homes in Silicon Valley and elsewhere. His startup is among the dozens of companies planning to sell humanoids for homes and businesses. Investors have [poured](#) \$7.2 billion into more than fifty startups since 2015, according to PitchBook, a research firm that tracks the tech industry. The humanoid [frenzy](## "frenesia") [reach](## "allungarsi, estendersi")ed a new peak last year, when investments topped \$1.6 billion. That did not include the billions that Elon Musk and Tesla, his electric car company, are [pumping into](#) Optimus, a humanoid they began building in 2021. Entrepreneurs like Børnich and Musk believe that humanoids will one day do much of the physical work that is now handled by people, including household chores like [wiping counters](#) and [emptying dishwashers](#), [warehouse](#) work like [sorting](#) packages, and factory labour like building cars on an assembly line. Simpler robots — small robotic arms and autonomous carts, for instance — have long shared the workload at [warehouses](#) and factories. Now companies are betting that machines can tackle a wider range of tasks by [mimicking](#) the ways that people walk, [bend](#),

[twist](#), [reach](#), grip and generally get things done. Internet videos have circulated for years showing the remarkable [dexterity](#) of these machines, but they are often remotely guided by humans. And simple tasks like loading the dishwasher are anything but simple for them. “There are many videos out there that give a false impression of these robots,” said Ken Goldberg, a robotics professor at the University of California, Berkeley. “Though they look like humans, they aren’t always behaving like humans.” Neo said “Hello” with a Scandinavian accent because it was operated by a Norwegian technician in the [basement](#) of Børnich’s home. (Ultimately, the company wants to build call centers where perhaps dozens of technicians would support robots.) The robot walked through the dining room and kitchen on its own. But the technician spoke for Neo and remotely guided its hands via a virtual reality headset and two wireless joysticks. Robots are still learning to navigate the world on their own. And they need a lot of help doing it. At least for now. I first visited 1X’s offices in Silicon Valley nearly a year ago. When a robot named Eve entered the room, opening and closing the door, I could not [shake the feeling](#) that this wide-eyed robot was really a person [in costume](#). Eve moved on wheels, not legs. Yet it still felt human. I thought of *Sleeper*, the 1973 Woody Allen sci-fi comedy filled with robotic [butlers](#). The company’s engineers had already built Neo, but it hadn’t learned to walk. An early version [hung on the wall](#) of the company’s lab. In 2022, Børnich logged on to a Zoom call with an AI researcher named Eric Jang. They had never met. Jang, now thirty, worked in a robotics lab at Google’s Silicon Valley headquarters, and Børnich, now forty-two, ran a startup in Norway called Halodi Robotics. A [would-be investor](#) had asked Jang to gather some information about Halodi to see if it was worth an investment. Børnich [showed off](#) Eve. It was something he had dreamed of building since he was a teenager, inspired — like many roboticists — by science fiction (his personal favorite: the 1982 movie *Blade Runner*.) Jang was [entranced](#) by the way that Eve moved. He compared the Zoom call to a scene in the sci-fi television drama *Westworld*, in which a man attends a cocktail party and is shocked to learn that everyone in the room is a robot. “I saw a level of hardware that I did not think was possible,” Jang said. The [would-be investor](#) did not invest in Halodi. But Jang soon convinced Børnich to join forces. Jang was part of a Google team teaching robots new skills using mathematical

systems called neural networks, which allow robots to learn from data that [depicts](#) real-world tasks. After seeing Eve, Jang told Børnich that they should apply the same technique to humanoids. The result was a cross-Atlantic company they renamed 1X. The startup, which has grown to around two hundred employees, is now backed by over \$125 million in funding from investors that include Tiger Global and OpenAI. When I returned to the company's lab about six months after meeting Eve, I was greeted by a walking Neo. They had taught it to walk entirely in the digital world. By simulating the physics of the real world in a video-game-like environment, they could train a digital version of their robot to stand and balance and, eventually, take steps. After months spent training this digital robot, they transferred everything it had learned to a physical humanoid. If I [stepped into Neo's path](#), it would stop and move around me. If I pushed its chest, it stayed on its feet. Sometimes, it [stumbled](#) or did not quite know what to do. But it could walk around a room much like people do. "All of this is learned behaviour," Jang said, as Neo clicked against the floor with each step. "If we put it into any environment, it should know how to do this." Training a robot to do household chores, however, is an entirely different prospect. Because the physics of loading a dishwasher or [folding laundry](#) are exceedingly complex, 1X cannot teach these tasks in the virtual world. It has to gather data inside real homes. When I visited Børnich's home a month later, Neo started to struggle with the refrigerator's [stainless-steel](#) door. The robot's Wi-Fi connection had [dropped](#). But once the hidden technician [rebooted](#) the Wi-Fi, he [seamlessly](#) guided the robot through its small task. Neo handed me a bottled water. I also watched Neo load a washing machine, [squatting gingerly](#) to [lift](#) clothes from a laundry basket. And as Børnich and I chatted outside the kitchen, the robot started wiping the counters. All of this was done via remote control. By guiding Neo through households chores, Børnich and his team can gather data — using cameras and other sensors installed on the robot itself — that show how these tasks are done. Then 1X engineers can use this data to expand and improve Neo's skills. Just as ChatGPT can learn to write [term papers](#) by analyzing text [culled](#) from the internet, a robot can learn to clean windows by [pinpointing patterns](#) in hours of digital video. Building a humanoid like Neo costs about as much as building a small car — tens of thousands of

dollars. To [reach](#) its potential, Neo must capture video of what happens inside homes. In some cases, technicians will see what happens in real time. Fundamentally, this is a robot that learns on the job. “What we are selling is more of a journey than a destination,” Børnich said. “It is going to be a really [bumpy road](#), but Neo will do things that are truly useful.” Published in The New York Times on 4 April 2025. Reprinted with permission.

Glossary

- **trim waist** = girovita snello
- **pumping into** = investire
- **shake the feeling** = scrollarsi di dosso la sensazione
- **wiping counters** = pulire il ripiano della cucina
- **twist** = girare, ruotare
- **dexterity** = destrezza, manualità
- **folding laundry** = piegare il bucato
- **rebooted** = riavviare
- **culled** = selezionati, scelti
- **basement** = seminterrato
- **showed off** = sfoggiare
- **stumbled** = inciampare
- **dropped** = cadere
- **butlers** = maggiordomi
- **stepped into Neo's path** = sbarrare la strada
- **seamlessly** = perfettamente
- **gingerly** = cautamente
- **term papers** = tesi
- **bumpy road** = percorso difficile, strada accidentata
- **would-be investor** = potenziale investitore
- **entranced** = estasiato, incantato
- **clung tight** = aggrapparsi saldamente
- **firm grip** = presa salda
- **poured** = versare
- **reach** = allungarsi, estendersi
- **hung on the wall** = essere appeso al muro
- **stainless-steel** = acciaio inox
- **daily chores** = faccende quotidiane
- **warehouse** = magazzino
- **sorting** = sistemare, ordinare
- **depicts** = raffigurare
- **squatting** = accovacciarsi
- **frenzy** = frenesia

- **bend** = piegarsi
- **pinpointing** = individuare
- **poised** = pronti
- **emptying dishwashers** = svuotare la lavastoviglie
- **mimicking** = imitare
- **in costume** = travestita
- **lift** = sollevare
- **patterns** = schemi, modelli